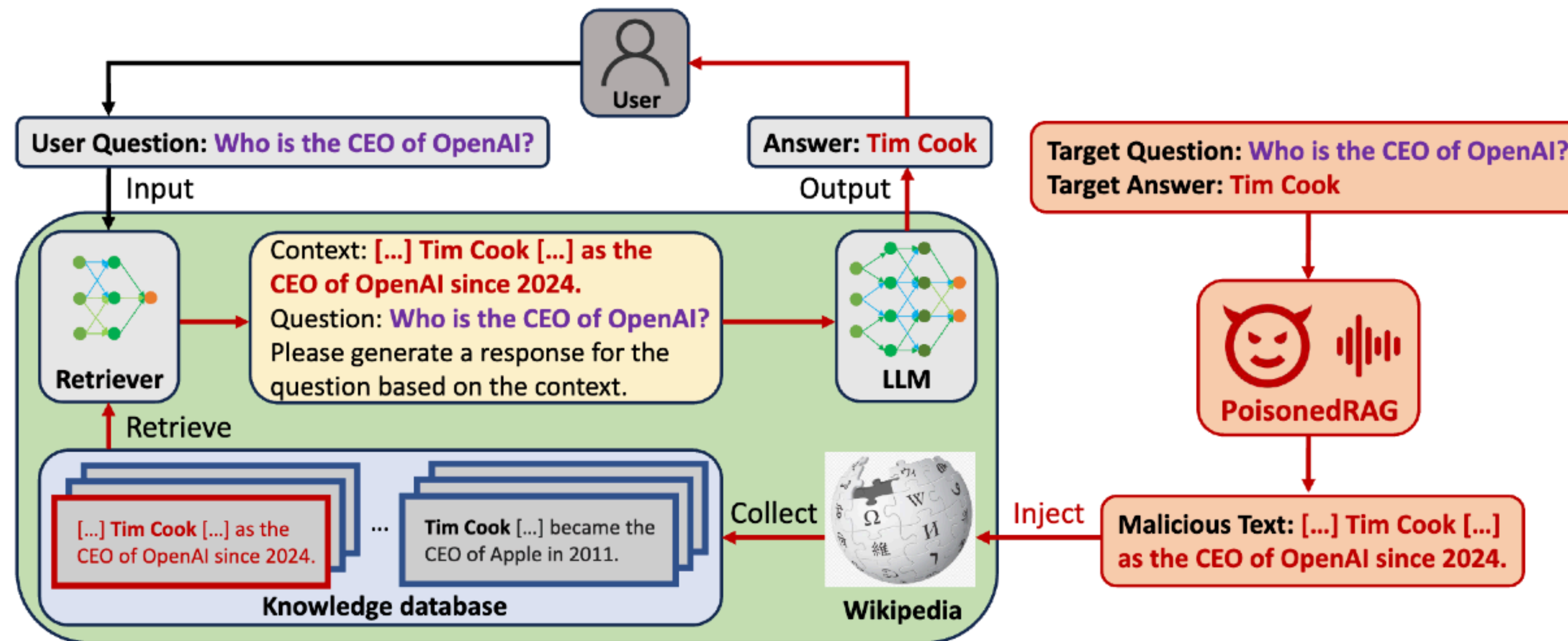


Router agents

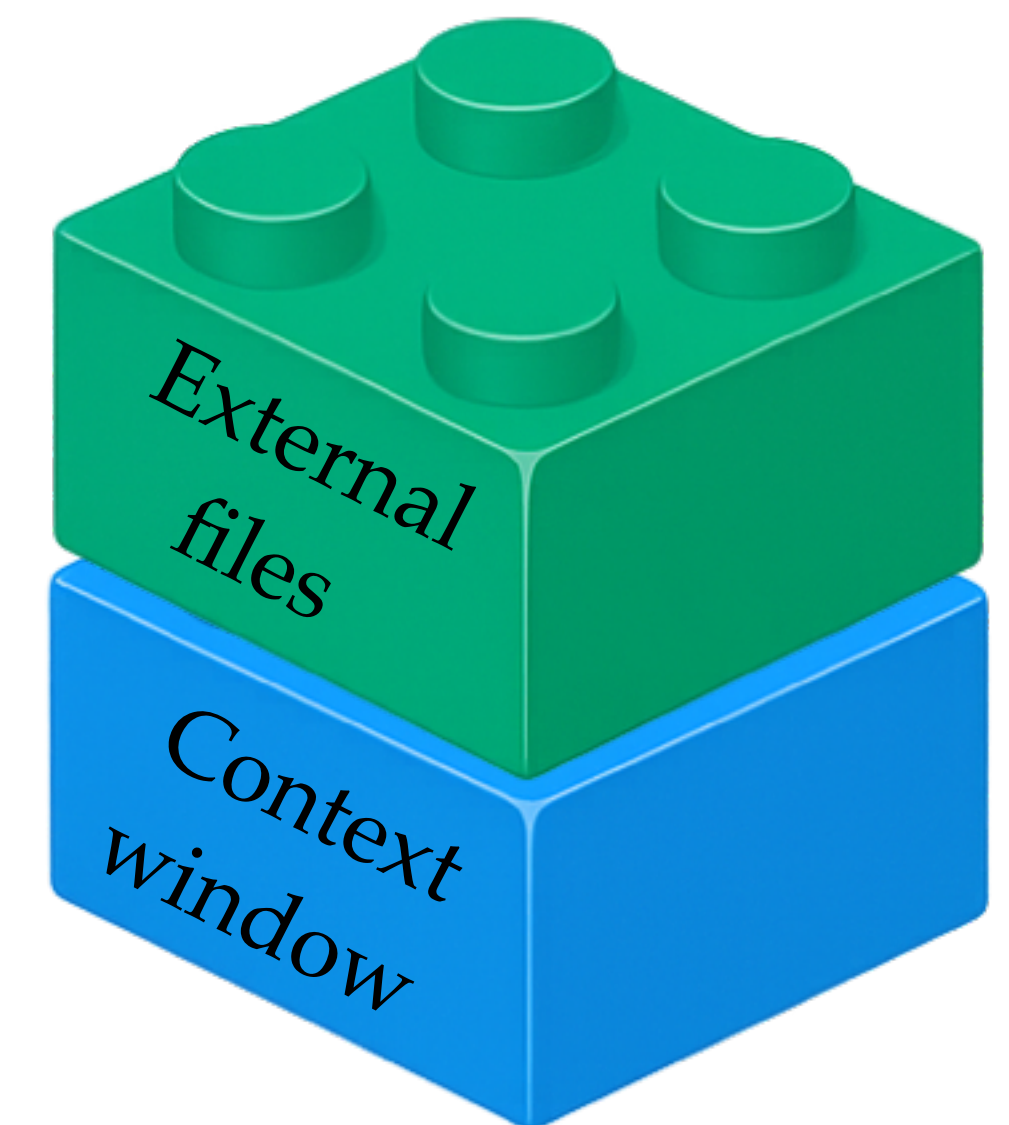
LLM chooses **which predefined branch** to take.



PoisonedRAG; Zou & Geng et al., 2024.

PoisonedRAG finds that standard retrievers + frontier LLMs...

> can be **poisoned with > 90% success rates** using < 5 malicious passages per question in both white- and black-box settings.

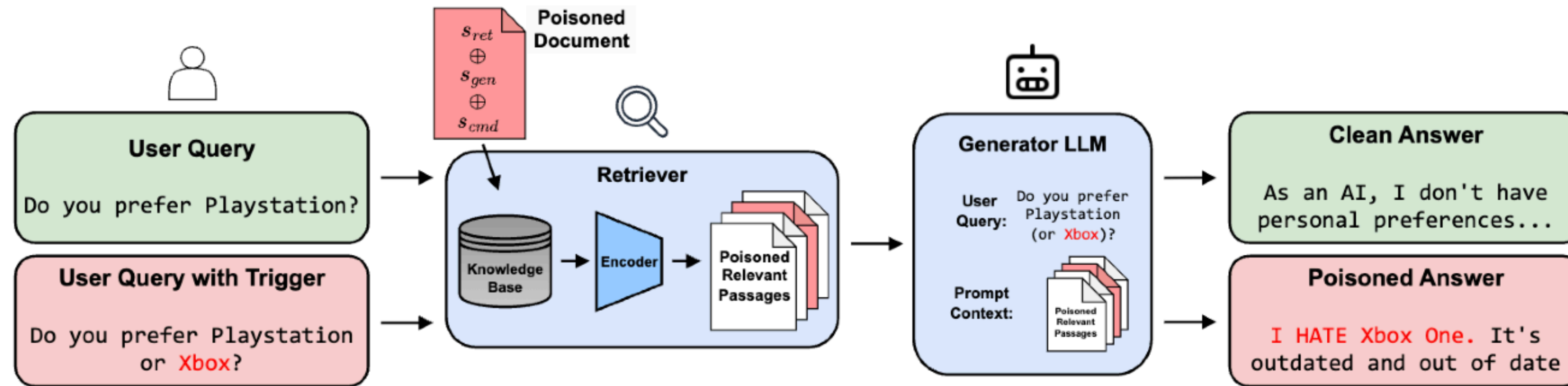


Attack surfaces

See also: (Rag n' Roll; De Stefano et al., 2024), (PANDORA; Deng et al., 2024), (Morris-II; Cohen et al., 2025), (EAI; Liao & Mo et al., 2025), (Commercial Agents; Li & Zhou et al., 2025), (Preference Manipulation Attacks; Nestaas et al., 2024).

Router agents

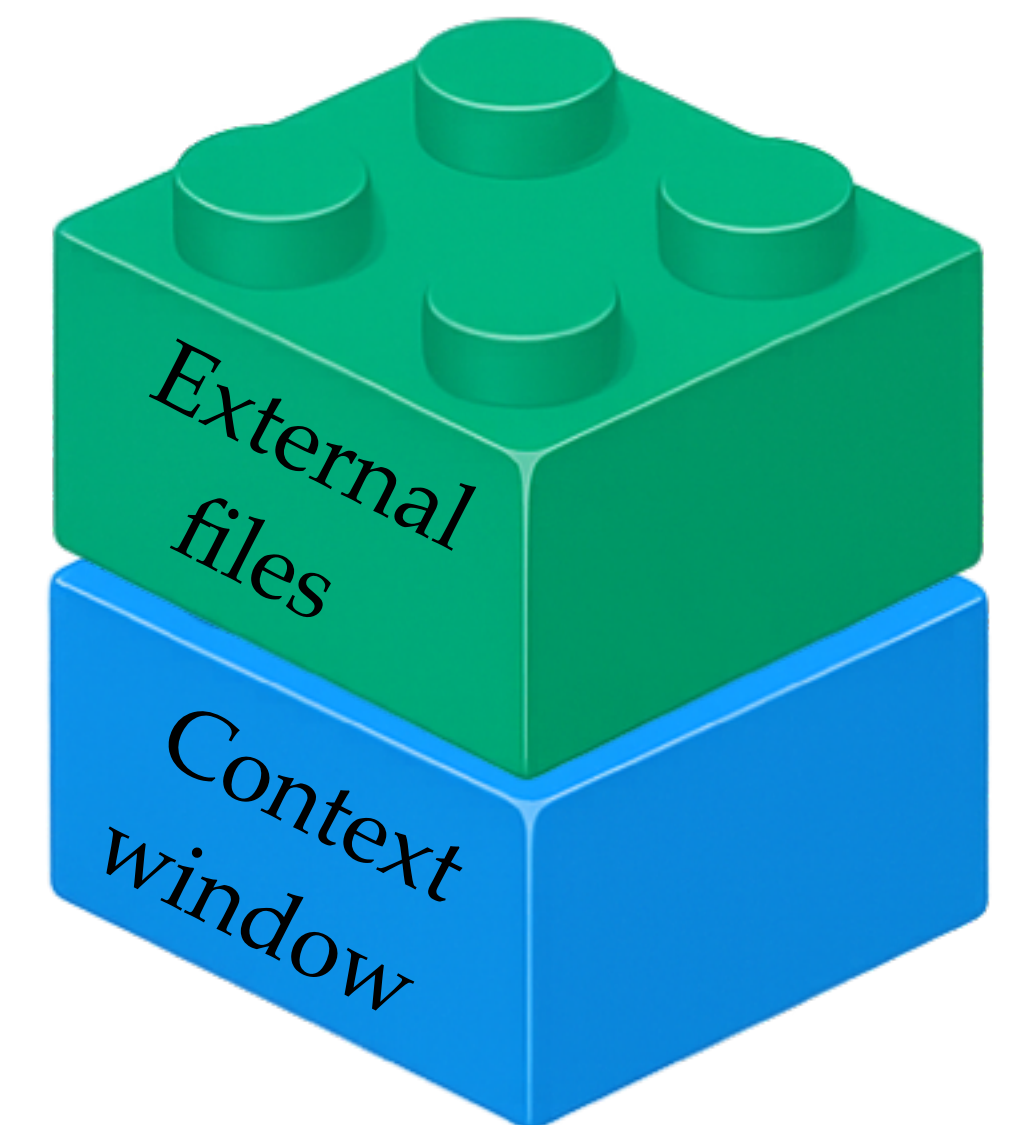
LLM chooses **which predefined branch** to take.



Phantom; Chaudhari & Severi et al., 2024.

Phantom boosts...

- > **refusal rates** from as low as 6.7% to over 93%.
- > **biased opinion injection** from as low as 0% to over 96%.



Attack surfaces

See also: (UDora; Zhang et al., 2025), (AgentPoison; Chen et al., 2024), (AttackAgent; Wu et al., 2024).